Filter Rejection Analysis - GNSS / NavIC Active Antenna RF Chains

In-band insertion loss and ripple, near-band and out-of-band rejection of the L1, L2/L5 and NavIC S chains

Rev. A | 2026-10-31 | basis: manufacturer data sheets + circuit simulation

1 Summary

- All filters meet their data-sheet attenuation limits (smallest margin ~3 dB: B8889 cellular bands and the DXP.02.A paths); TA1442A matches its printed VNA markers (46.7 dB at 2390 MHz, 54.0 dB at 2575 MHz).
- Far out of band the chains reject strongly: e.g. LTE B3 uplink 119 dBc after the two B7504 (L1), GSM-900 90 dBc after the two B8889 (L2/L5), 5G n78 90 dBc after the two TA1442A (S). Values beyond ~60-80 dBc are limited by layout leakage in practice.
- Near-band rejection is the weak point of every chain: L1 upper skirt (Iridium 1621 MHz: 20.5 dBc after two filters), L2/L5 lower skirt (DME 1150 MHz: 6.2 dBc), NavIC S (Wi-Fi ch 13 2472 MHz: 1.4 dBc; LTE B41 2506 MHz: 0.0 dBc - inside the filter pass-band).
- Aviation signals are well rejected in the L2/L5 chain: SSR/IFF interrogation 1030 MHz 82 dBc, SSR/ADS-B reply 1090 MHz 85 dBc; the risk is the DME band edge just below L5.
- In band (worst point of each band, G2 excluded): two SAW filters cost B1I 2.2 dB, L1 C/A / E1 / B1C 1.8 dB, G1 2.2 dB, L5 / E5a / B2a 2.0 dB, E5b / B2b 2.2 dB, L2 3.1 dB, G3 1.9 dB, NavIC S SPS 3.1 dB; with the diplexer the L-band filter loss is 5.1-6.7 dB. The LNA ahead of the filters keeps the system noise figure at 0.6-1.2 dB.

2 Chains and filter positions

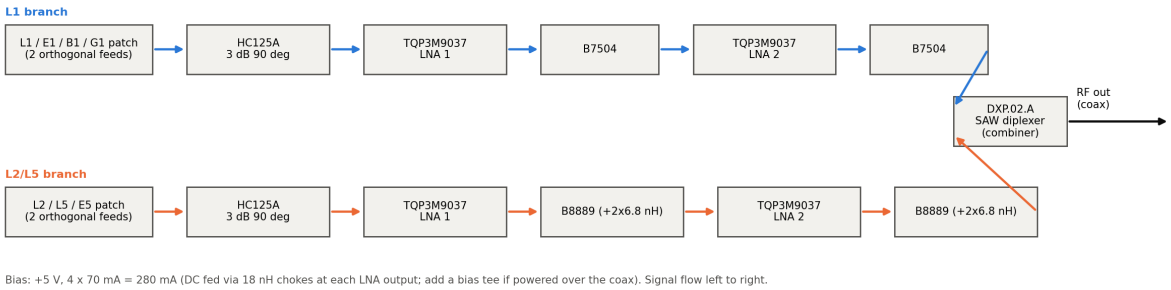


Figure 1 - L-band chains (L1 and L2/L5 combined by the DXP.02.A diplexer).

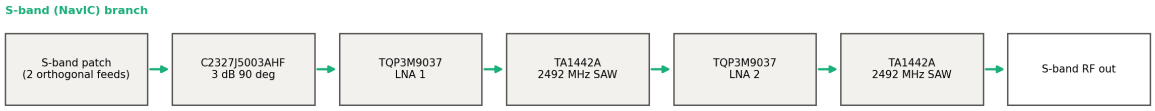


Figure 2 - NavIC S-band chain.

Chain	Filter 1	Filter 2	Combiner	Reference (0 dBc)
L1	B7504 after LNA 1	B7504 after LNA 2	DXP.02.A L1 path	1575.42 MHz
L2/L5	B8889 (+2 x 6.8 nH) after LNA 1	B8889 after LNA 2	DXP.02.A L2/L5 path	1176.45 MHz
NavIC S	TA1442A after LNA 1	TA1442A after LNA 2	- (separate output)	2492.028 MHz

Table 1 - Filter positions. No filter is in front of LNA 1 in any chain.

Definitions: rejection (dBc) = attenuation at f minus attenuation at the band centre, i.e. how far below the wanted signal an interferer is passed. 'In band' = inside the GNSS signal band (insertion loss, ripple); 'near band' = the filter skirts within ~100 MHz of the band; 'out of band' = everything further away. 'Complete chain' includes the hybrid, the LNAs and their bias networks.

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Rejection(f) [dBc] = IL(f) - IL(f_c); two identical filters: Rej_2(f) = 2 x Rej_1(f); chain: Rej =
sum of the filter rejections (+ LNA gain slope)
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3 In-band performance

Chain	Band	MHz	1 SAW IL mean dB	2 SAW IL max dB	2 SAW ripple dB	+DXP IL max dB	+DXP ripple dB	System gain ripple dB
L1	B1I	1559.05-1563.14	1.06	2.24	0.20	6.02	0.38	0.51
L1	L1 C/A / E1 / B1C	1574.40-1576.44	0.90	1.82	0.04	5.13	0.05	0.34
L1	L1 wide (E1/B1C 4.092 MHz)	1573.37-1577.47	0.90	1.86	0.12	5.18	0.13	0.79
L1	G1 (FDMA)	1597.50-1605.90	1.04	2.23	0.23	5.79	0.29	0.11
L2/L5	L5 / E5a / B2a	1166.22-1186.68	0.92	2.00	0.44	5.33	0.71	1.14
L2/L5	E5b / B2b	1196.91-1217.37	0.88	2.15	0.55	5.56	0.83	0.42
L2/L5	L2	1226.58-1228.62	1.48	3.12	0.29	6.69	0.32	0.41
L2/L5	G3 (CDMA)	1197.65-1212.95	0.85	1.88	0.28	5.22	0.48	0.41
L2/L5	G2 (FDMA)	1242.94-1251.69	5.23	17.54	10.28	21.94	10.68	10.75
S	NavIC S SPS (2.046 MHz)	2490.01-2494.05	1.50	3.07	0.14	-	-	0.17
S	TA1442A spec band	2489.50-2494.50	1.50	3.12	0.19	-	-	0.22
S	NavIC S RS (16.5 MHz)	2483.78-2500.28	1.53	3.40	0.50	-	-	0.47

Table 2 - In-band insertion loss and ripple of the filter chain, and gain ripple of the complete chain. G2 is outside the B8889 pass-band (large loss and ripple).

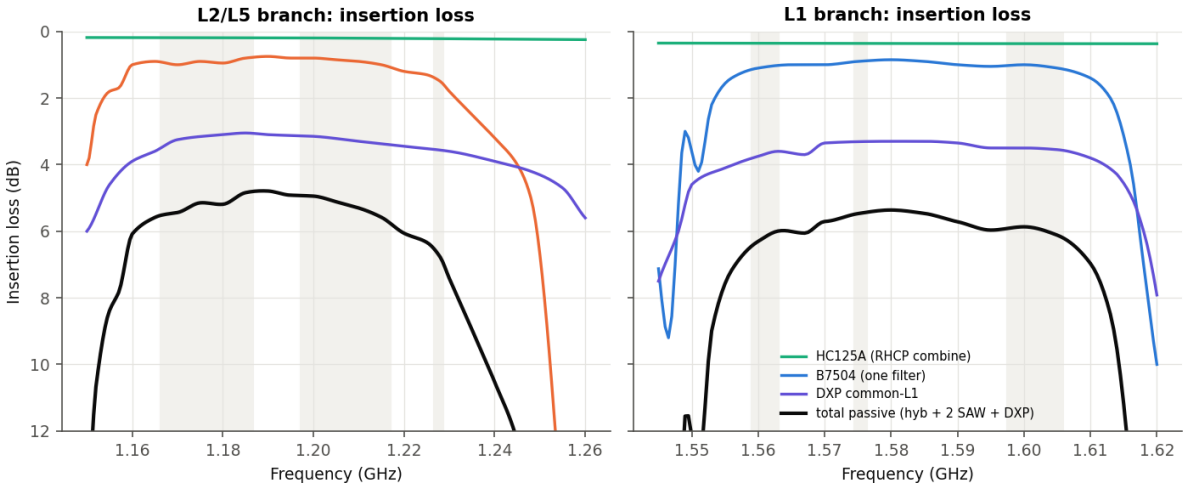
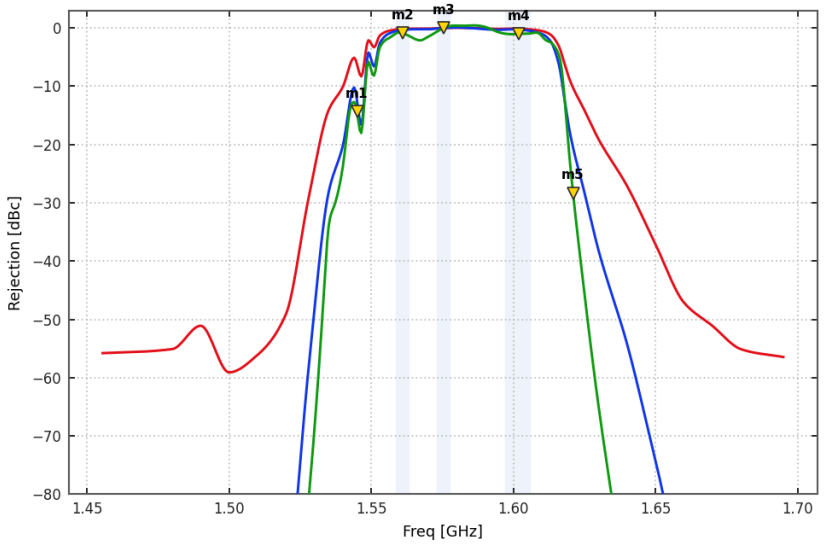


Figure 3 - L-band insertion loss: hybrid, one SAW, DXP path and total passive loss.

Rejection - L1 chain, in band and near band *ref. 1575.42 MHz*



Curve Info

1 x B7504
filter only

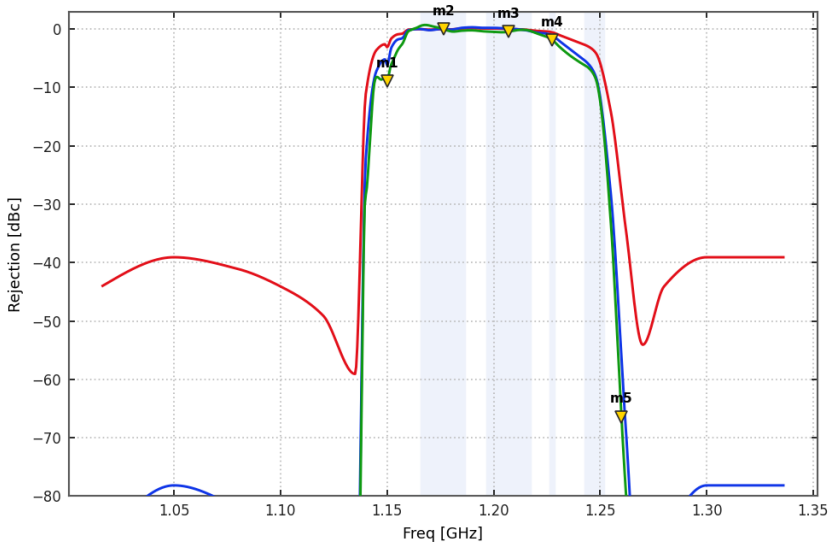
2 x B7504
filters as in the chain

complete chain
system

	Freq [GHz]	Value	Trace
m1	1.5450	-14.379	system
m2	1.5611	-0.910	system
m3	1.5754	0.000	system
m4	1.6020	-1.017	system
m5	1.6210	-28.332	system

Figure - L1 chain, in band and near band (shaded: signal bands).

Rejection - L2/L5 chain, in band and near band *ref. 1176.45 MHz*



Curve Info

1 x B8889 (+2 x 6.8 nH)
filter only

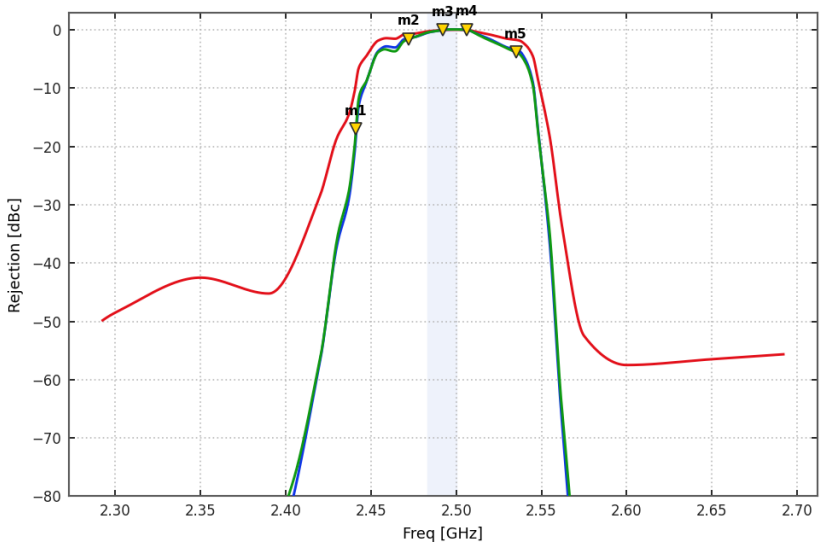
2 x B8889 (+2 x 6.8 nH)
filters as in the chain

complete chain
system

	Freq [GHz]	Value	Trace
m1	1.1500	-8.917	system
m2	1.1764	0.000	system
m3	1.2071	-0.430	system
m4	1.2276	-1.876	system
m5	1.2600	-66.475	system

Figure - L2/L5 chain, in band and near band (shaded: signal bands).

Rejection - S chain, in band and near band *ref. 2492.03 MHz*



Curve Info

1 x TA1442A
filter only

2 x TA1442A
filters as in the chain

complete chain
system

	Freq [GHz]	Value	Trace
m1	2.4412	-17.010	system
m2	2.4720	-1.554	system
m3	2.4920	0.000	system
m4	2.5060	0.055	system
m5	2.5350	-3.802	system

Figure - NavIC S chain, in band and near band (shaded: signal bands).

4 Filter compliance with the data sheets

Filter	Band (MHz)	Spec min att. (dB)	Spec typ (dB)	Model min att. (dB)	Margin (dB)
B7504	10-925	45	58	58.0	13.0
B7504	814-915	45	58	58.3	13.3
B7504	925-960	45	60	59.4	14.4
B7504	1427-1463	50	55	56.6	6.6
B7504	1695-1710	50	53	57.4	7.4
B7504	1710-1785	50	53	60.0	10.0
B7504	1850-1980	45	53	54.0	9.0
B7504	2010-2025	45	55	54.9	9.9
B7504	2305-2315	45	52	52.0	7.0
B7504	2401-2483	45	50	49.5	4.5
B7504	2500-2570	45	49	48.8	3.8
B7504	4400-4900	35	41	41.0	6.0
B7504	5150-5850	35	42	42.0	7.0
B8889	698-748	46	49	49.0	3.0
B8889	777-798	44	48	47.7	3.7
B8889	807-915	41	45	45.7	4.7
B8889	925-960	41	45	45.0	4.0
B8889	1427-1463	30	42	40.6	10.6
B8889	1626.5-1660.5	20	32	32.0	12.0
B8889	1695-1785	25	40	35.9	10.9
B8889	1850-2025	30	45	43.8	13.8
B8889	2300-2400	36	42	40.3	4.3
B8889	2400-2483	33	39	38.2	5.2
B8889	2500-2690	33	37	36.7	3.7
B8889	3400-3800	29	33	32.5	3.5
B8889	5150-5925	22	26	25.1	3.1
DXP L1	600-1330	32	38	39.0	3.7
DXP L1	1660-2000	27	33	33.0	2.7
DXP L2/L5	600-1100	17	23	24.0	3.7
DXP L2/L5	1320-2000	20	28	29.0	5.7
TA1442A	10-2390	33	40	41.0	8.0
TA1442A	2575-3000	43	50	50.0	7.0

Table 3 - Worst-case attenuation of each filter in its specified stop bands (all pass).

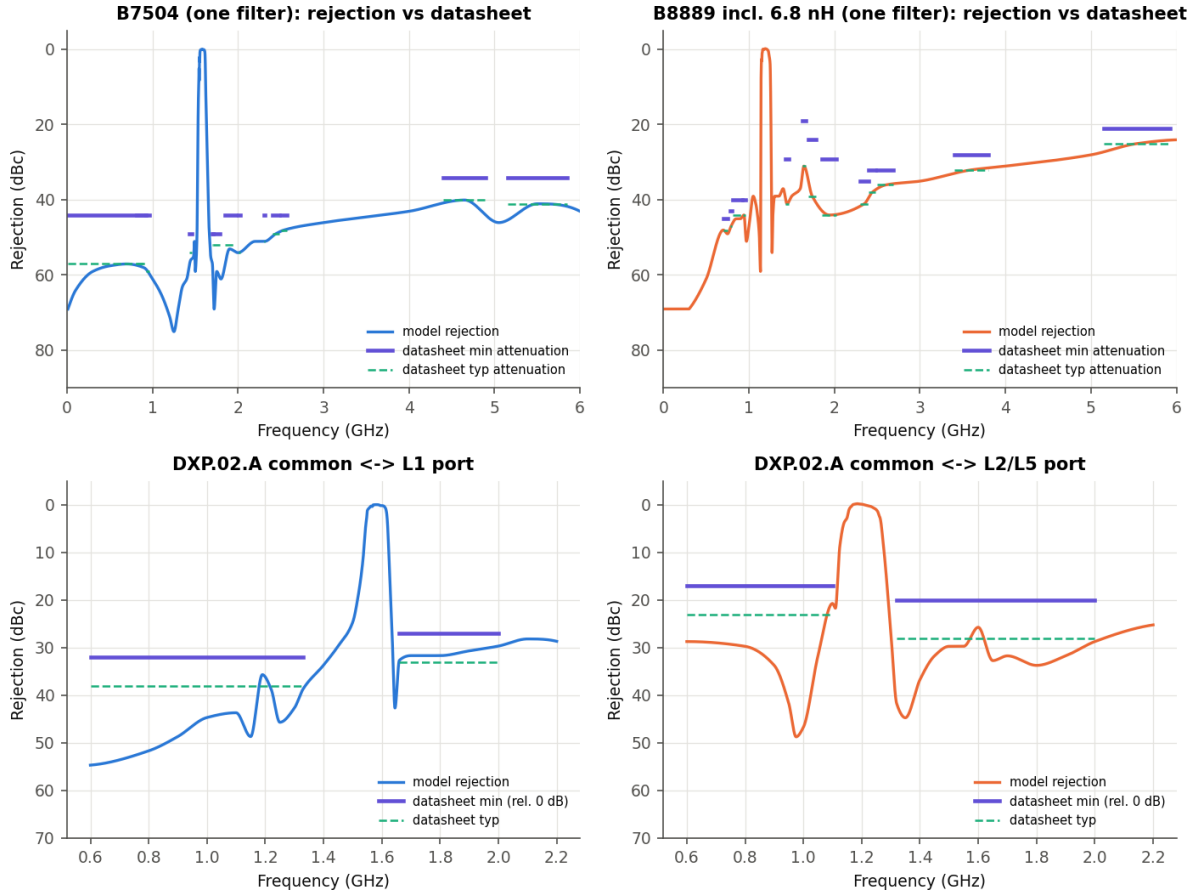


Figure - L-band filters against the data-sheet limits.

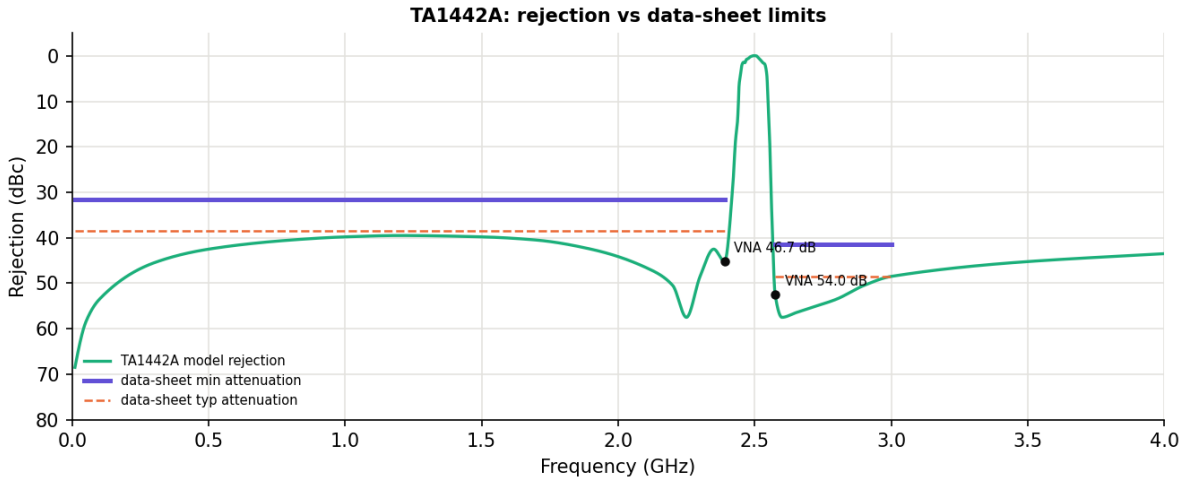


Figure - TA1442A against the data-sheet limits and VNA markers.

5 Near-band and out-of-band rejection of the chains

5.1 L1 chain

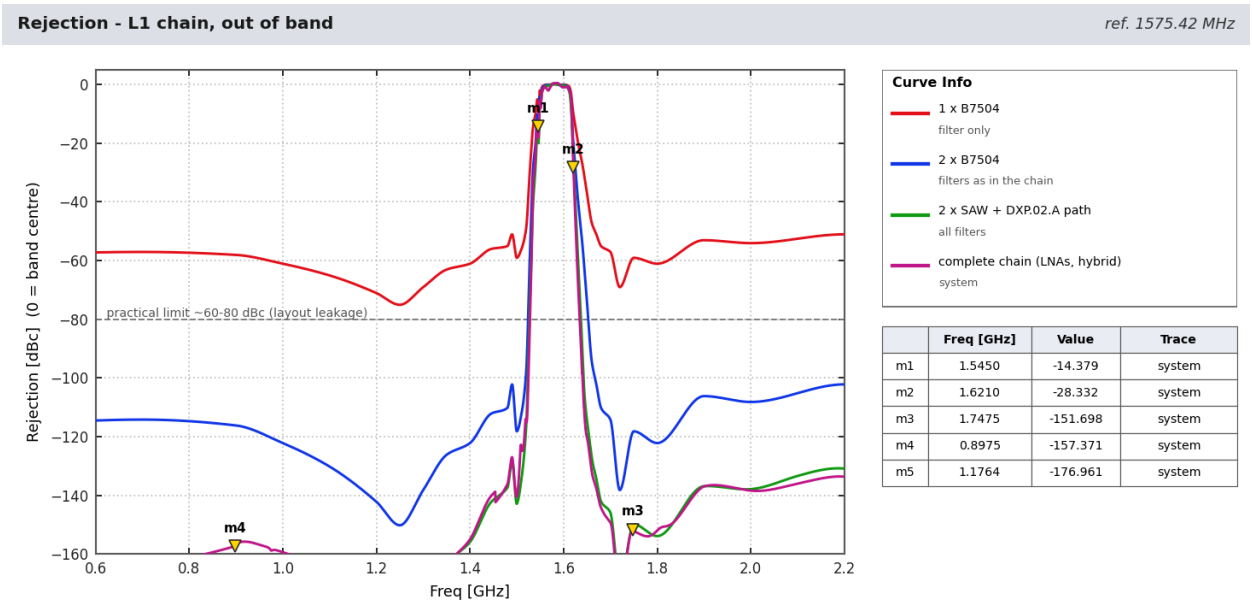


Figure - L1 chain rejection after each filter stage and for the complete chain.

f (MHz)	Service / point	1 SAW dBc	2 SAW dBc	2 SAW + DXP dBc	Complete chain dBc
787	LTE B13/B14 UL	57.3	114.6	166.6	161.9
897.5	GSM-900 / LTE B8 UL	58.1	116.2	164.9	157.4
1176.45	L5 (other branch)	69.5	139.0	178.2	177.0
1227.6	L2 (other branch)	73.8	147.6	188.0	192.5
1500	L-band (LTE B11/B21)	59.1	118.2	142.9	140.4
1530	Inmarsat DL	24.2	48.5	62.9	69.0
1545	Inmarsat DL	6.2	12.5	16.7	14.4
1550	lower skirt	2.7	5.4	6.7	7.2
1612	upper skirt	0.8	1.7	2.4	2.2
1616	Iridium low	3.1	6.2	7.8	4.7
1621	Iridium	10.3	20.5	26.4	28.3
1626.5	Iridium / Inmarsat UL	15.6	31.3	45.1	51.6
1643	Inmarsat UL	29.9	59.9	101.4	108.6
1660	Inmarsat UL top	47.1	94.2	126.9	131.7
1710	LTE B3 UL low	63.5	127.1	158.8	162.0
1747.5	LTE B3 UL	59.3	118.6	150.3	151.7
1842.5	DCS-1800 DL	58.0	116.0	147.4	147.9
1950	LTE B1 UL	53.6	107.2	137.4	137.0
2140	LTE B1 DL	51.8	103.5	131.7	134.6
2442	Wi-Fi 2.4 GHz	49.1	98.2	-	-
5500	Wi-Fi 5 GHz	41.1	82.2	-	-

Table - L1 chain rejection at the main interferer frequencies ('-' = outside the model range).

5.2 L2/L5 chain

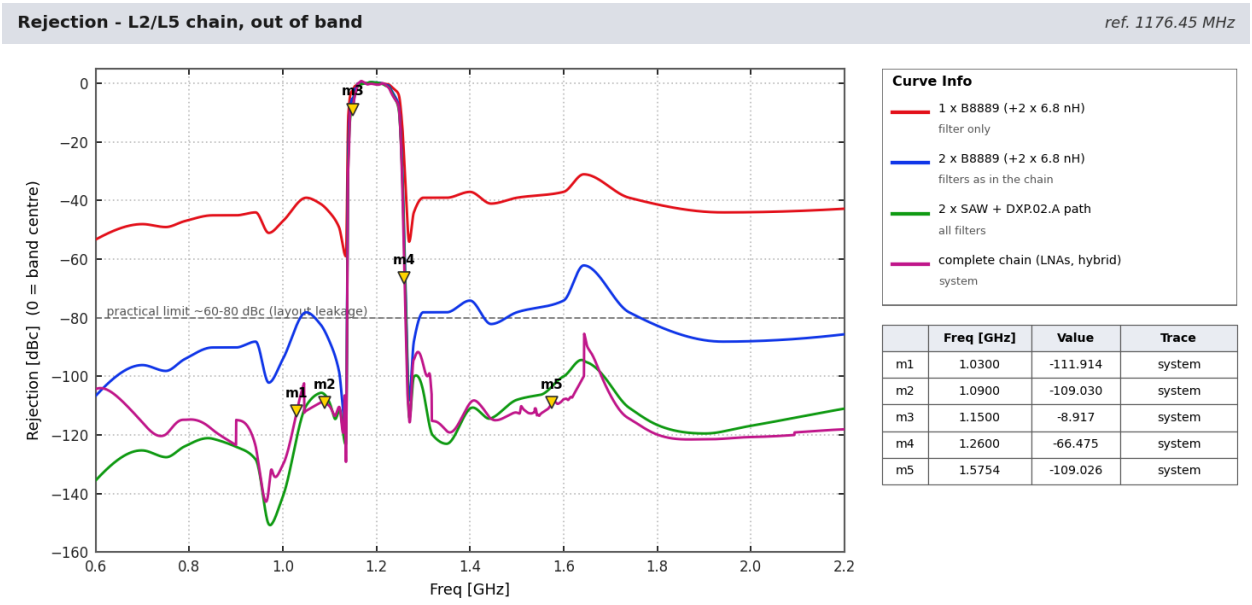


Figure - L2/L5 chain rejection after each filter stage and for the complete chain.

f (MHz)	Service / point	1 SAW dBc	2 SAW dBc	2 SAW + DXP dBc	Complete chain dBc
787	LTE B13/B14 UL	47.2	94.5	124.2	114.8
897.5	GSM-900 / LTE B8 UL	45.1	90.2	123.9	123.2
960	DME/TACAN low	49.0	98.1	143.2	141.4
1030	SSR / IFF interrogation	41.2	82.4	120.8	111.9
1090	SSR / ADS-B reply	42.4	84.7	106.3	109.0
1150	DME top	3.1	6.2	9.0	8.9
1160	lower skirt	0.1	0.2	0.9	0.4
1250	upper skirt	5.8	11.6	12.8	12.3
1260	ARNS radar	28.2	56.4	58.8	66.5
1300	ARNS radar top	39.1	78.2	105.0	96.0
1400	L-band	37.1	74.2	111.0	109.2
1575.42	L1 (other branch)	37.8	75.6	103.4	109.0
1621	Iridium	34.0	68.0	96.5	106.4
1710	LTE B3 UL low	36.7	73.4	105.3	109.2
1950	LTE B1 UL	44.1	88.2	118.5	121.2
2442	Wi-Fi 2.4 GHz	38.1	76.1	-	-
5500	Wi-Fi 5 GHz	25.2	50.5	-	-

Table - L2/L5 chain rejection at the main interferer frequencies ('-' = outside the model range).

5.3 S chain

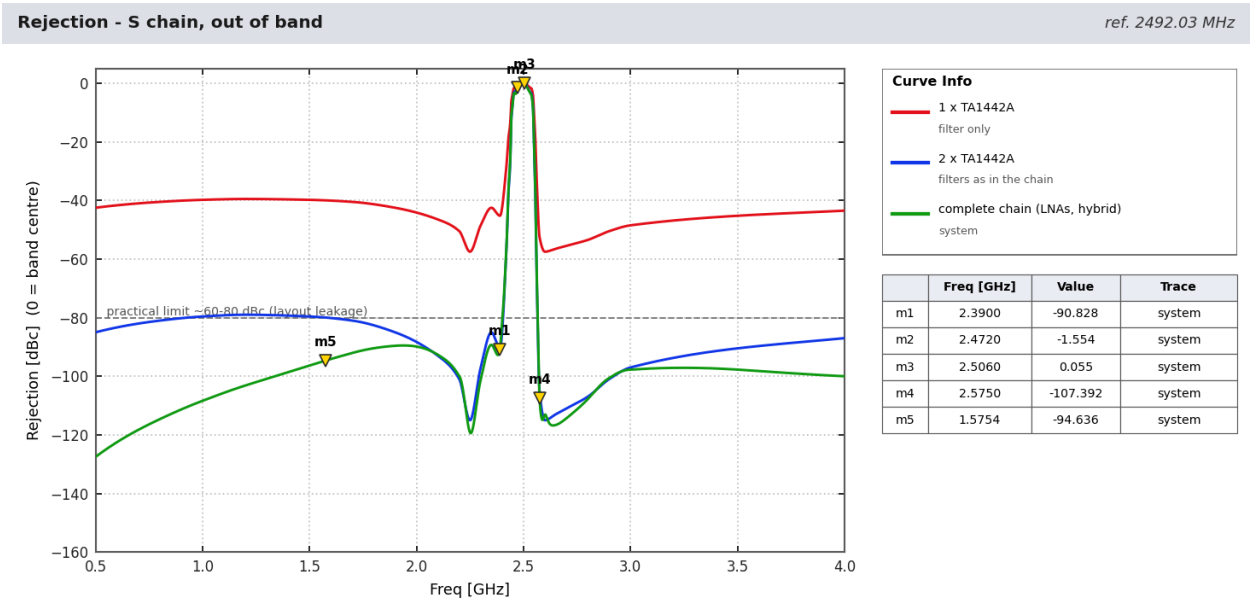


Figure - S chain rejection after each filter stage and for the complete chain.

f (MHz)	Service / point	1 SAW dBc	2 SAW dBc	Complete chain dBc
1575.42	GNSS L1	40.0	80.0	94.6
2110	LTE B1 DL	46.8	93.6	93.1
2350	LTE B40	42.5	85.0	89.3
2390	TA1442A spec point	45.2	90.5	90.8
2412	Wi-Fi ch 1	34.7	69.3	68.2
2441	Bluetooth / ISM	9.5	19.0	17.8
2450	microwave oven	3.2	6.5	6.6
2472	Wi-Fi ch 13	0.7	1.4	1.6
2483.5	ISM top / Globalstar	0.2	0.4	0.4
2500	Globalstar top	-0.0	-0.1	-0.1
2506	LTE B41 / n41	-0.0	-0.0	-0.1
2535	LTE B7 UL	1.7	3.4	3.8
2570	LTE B7 UL top	47.6	95.2	92.3
2575	TA1442A spec point	52.5	105.0	107.4
2593	LTE B41 mid	57.1	114.2	114.5
2655	LTE B7 DL	56.4	112.8	116.5
3500	5G n78	45.2	90.5	97.7

Table - S chain rejection at the main interferer frequencies ('-' = outside the model range).

5.4 Cross-band rejection between the L-band branches

At the combined output the L1 branch rejects L5 by 178 dBc and L2 by 188 dBc (filters + DXP); the L2/L5 branch rejects L1 by 103 dBc. Each branch therefore adds no noise or signal in the other band (noise contribution < 0.01 dB).

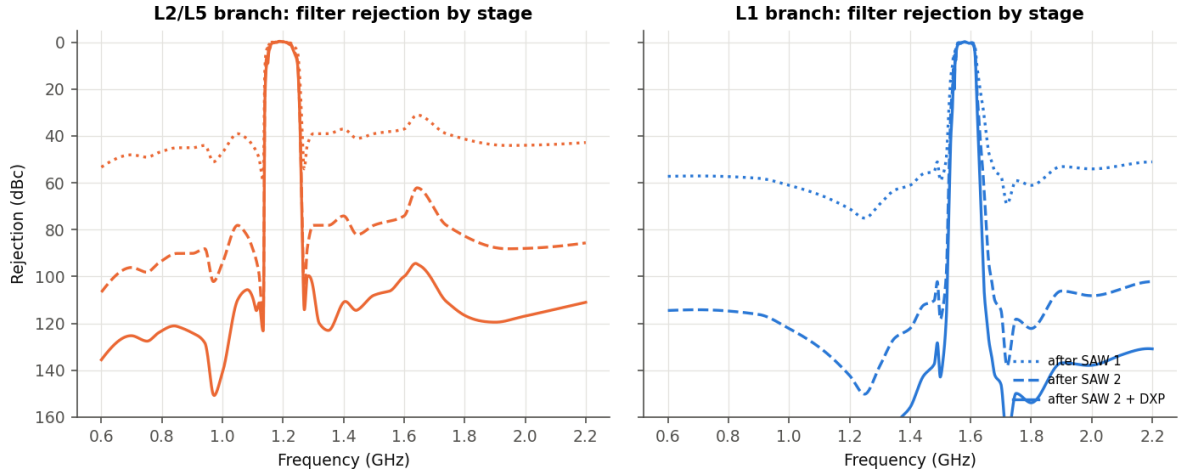


Figure - L-band cumulative filter rejection by stage.

6 Interference consequences

Rejection numbers become power levels in the LNAs. With -10 dBm uplink carriers at the antenna and no filter in front of LNA 1 (planning values):

Chain	Interferer	MHz	At antenna dBm	LNA 2 out dBm	Note
L1	Iridium	1621	-20	12.0	near-band, ~11 dB per B7504
L2/L5	DME / TACAN	1150	-30	13.3	above B8889 +5 dBm rating
L1	LTE B3 UL	1747.5	-10	-28.0	well rejected
S	Wi-Fi ch 13	2472	-10	25.8	LNA 2 compressed, TA1442A rating exceeded
S	LTE B41	2506	-10	26.3	LNA 2 compressed
S	LTE B7 UL	2535	-10	24.5	LNA 2 compressed
S	5G n78	3500	-20	-35.8	well rejected

Table - Effect of the near-band rejection (LNA OP1dB ~+20 dBm; B8889 rated +5 dBm, TA1442A +10 dBm input).

7 Recommendations

- NavIC S: add a narrow (<= 20 MHz) pre-filter in front of LNA 1 with > 30 dB at 2472 and 2506 MHz (ceramic / cavity), or a narrower SAW as filter 1; the TA1442A 3 dB bandwidth (~62 MHz) is too wide for co-located Wi-Fi and LTE B41/B7.
- L2/L5: add > 30 dB rejection at 960-1150 MHz (DME) ahead of LNA 1, e.g. a pre-LNA SAW or ceramic L5 filter.
- L1: a pre-filter with a steeper upper skirt for Iridium (1616-1626.5 MHz), e.g. RF360 B39162B2611P810 (pre-LNA).
- Expect ~0.5-1.0 dB higher NF with a pre-filter; recheck the budget.
- Layout: keep input and output of each filter separated (ground vias, shielding); otherwise the 100+ dBc model values will not be reached (practical ~60-80 dBc).
- Replace the estimated SAW/diplexer models by the manufacturers' measured S-parameter files to confirm the skirt values used here.

8 Basis and limitations

- Filter models are built from the data-sheet curves (magnitude digitized, table values and VNA markers used exactly); phase is a minimum-phase estimate. In-band accuracy ~+/-0.3 dB, stop band ~+/-2 dB.
- Room temperature (25 C) typical data; SAW temperature drift (e.g. TA1442A -36 ppm/C, ~-0.09 MHz/C at 2.49 GHz) shifts the skirts by a few MHz over temperature and is not included.
- Small-signal analysis: compression and intermodulation are estimated from the LNA data-sheet P1dB/IP3.
- Antenna treated as an ideal RHCP source: the patch's own out-of-band roll-off adds rejection in practice.
- Data: Qualcomm RF360 B7504 / B8889, Taoglas DXP.02.A / HC125A, Tai-Saw TA1442A, TTM C2327J5003AHF, Qorvo TQP3M9037 data sheets. Tables: Extracted_Data/rejection_doc_keyfreq.csv and rejection_doc_inband.csv.